Mittuniversitetet
MID SWEDEN UNIVERSITY

DT081A – Signal and Image Processing

Lecture 8 - Spatial domain image processing I

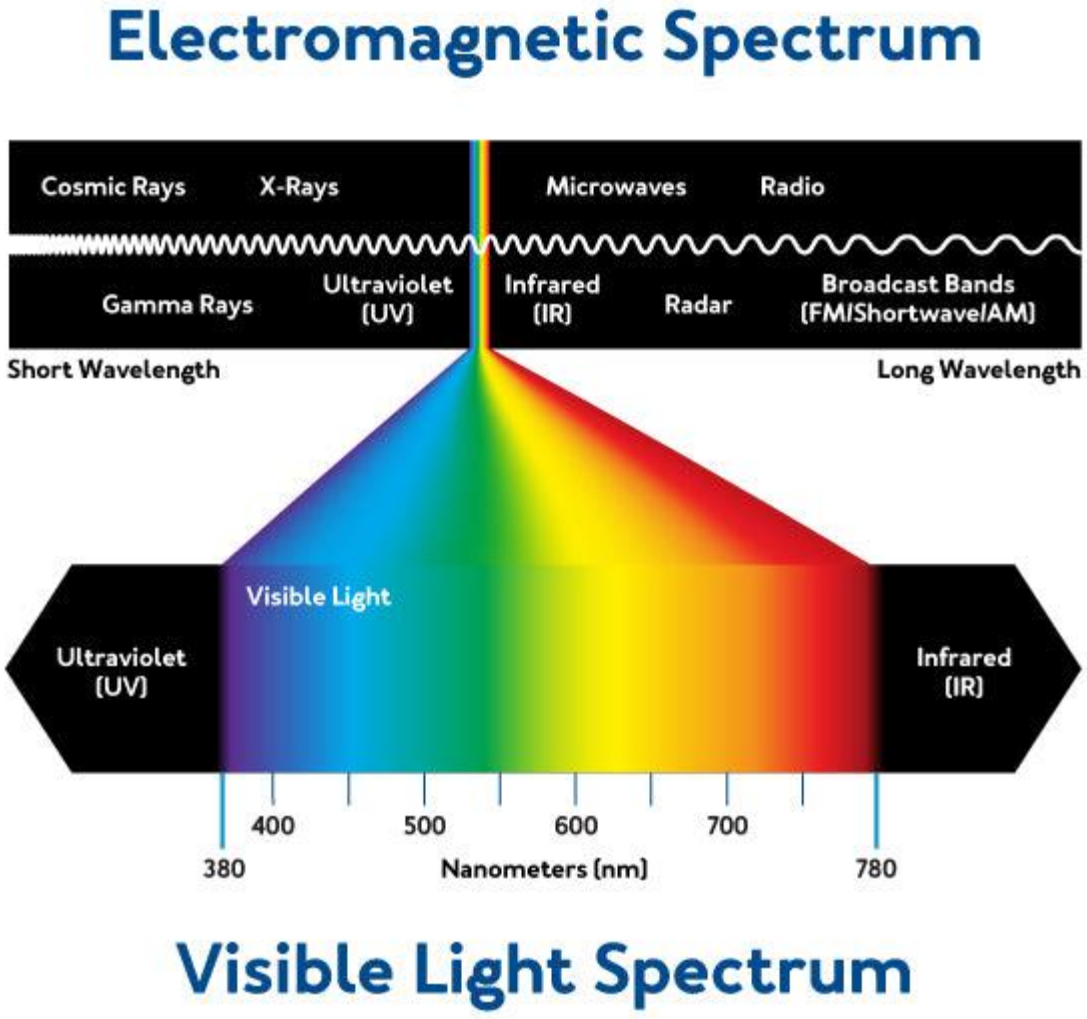


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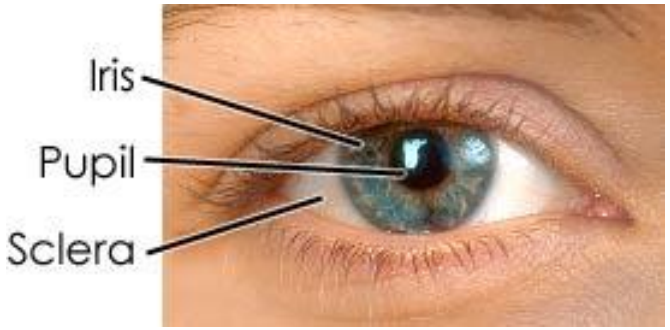
Light and Spectra

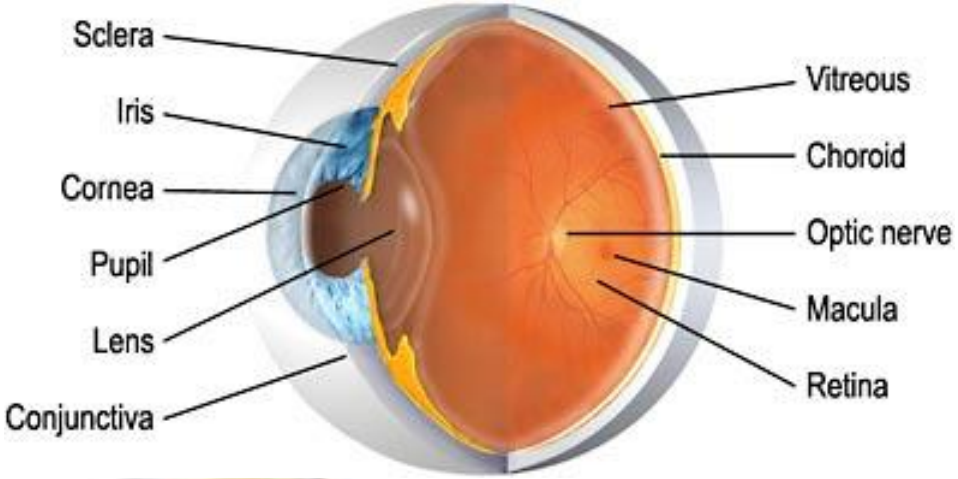
Light is an electromagnetic wave. Its color is characterized by the wavelength content of the light.

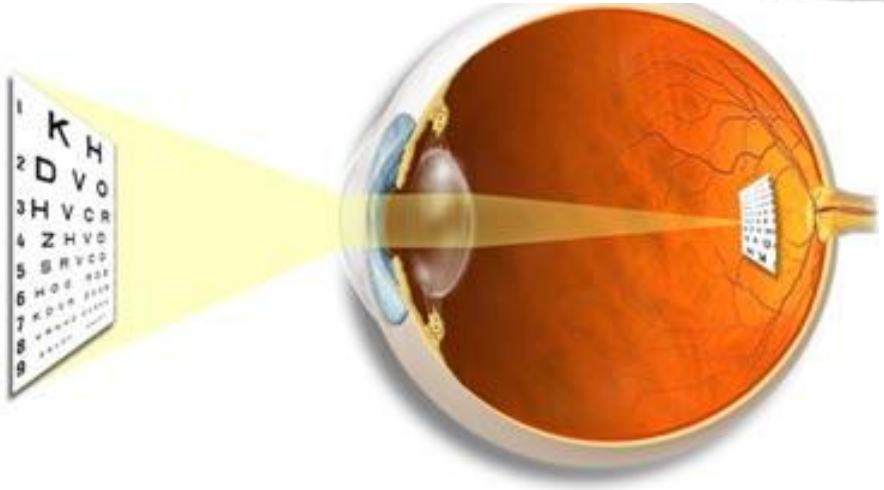
- Laser = single wavelength
- Most light sources = many contributing wavelengths
- Humans can only detect contributions falling within the visible wavelengths (in the range 400 nm to 700 nm)



Eye Anatomy



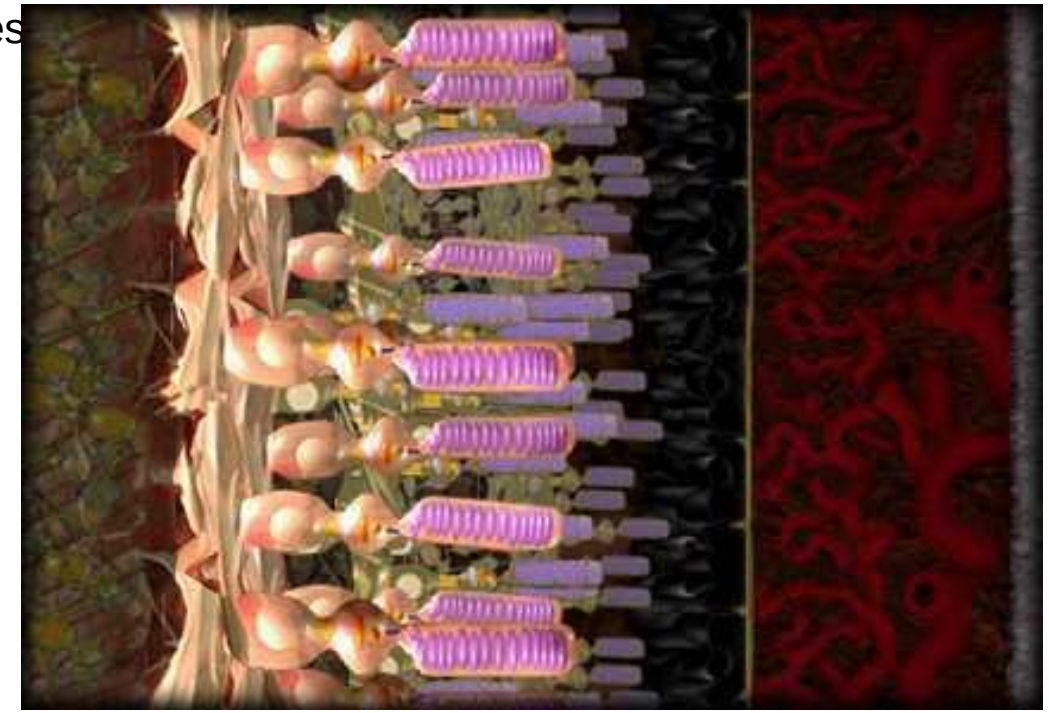




From http://www.stlukeseye.com/anatomy/

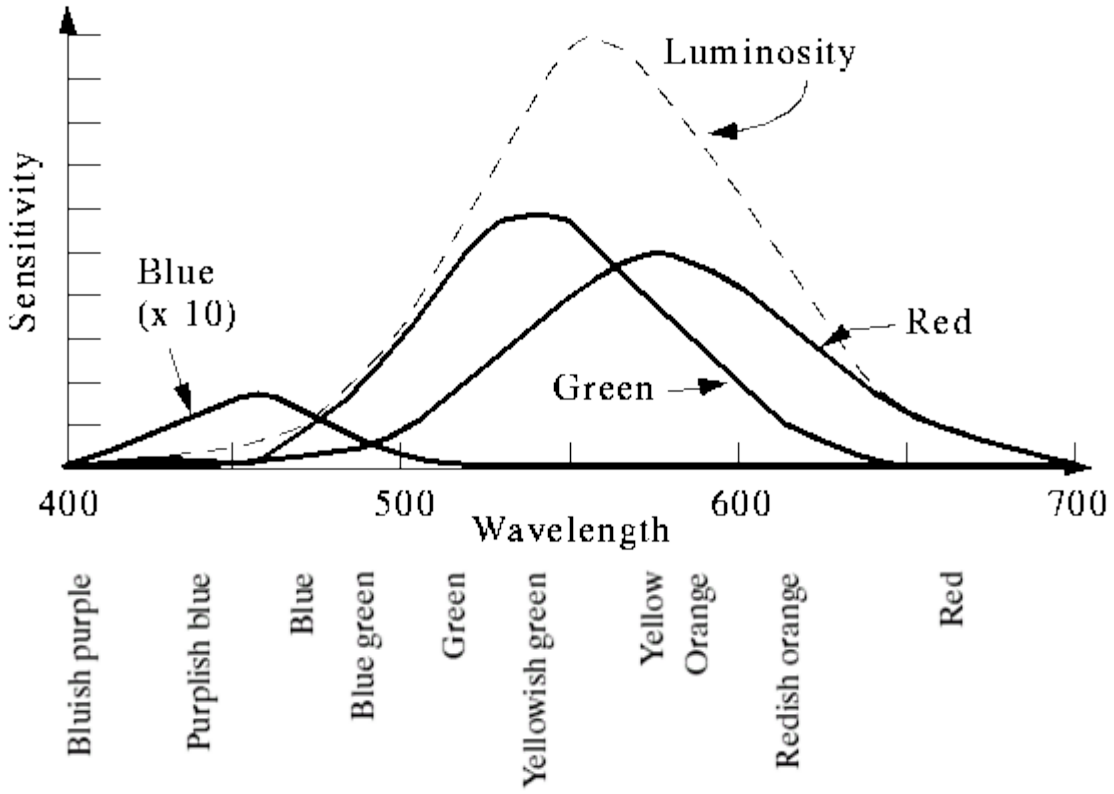
Human Perception of Color

- Retina contains photo receptors
  - Cones: day vision, can perceive color tone
    - Red, green, and blue cones
    - Different cones have different frequency responses
    - Tri-receptor theory of color vision [Young1802]
  - Rods: night vision, perceive brightness only
- Color sensation is characterized by
  - Luminance (brightness)
  - Chrominance
    - Hue (color tone)
    - Saturation (color purity)



Source: http://www.macula.org/anatomy/retinaframe.html

Frequency Responses of Cones and the Luminous Efficiency Function



$$C_i = \int C(\lambda) a_i(\lambda) d\lambda, \quad i = r, g, b, y$$

Illuminating and Reflecting Light

- Illuminating sources:
  - emit light (e.g. the sun, light bulb, TV monitors)
  - perceived color depends on the emitted freq.
  - follows additive rule
    - R+G+B=White
- Reflecting sources:
  - reflect an incoming light (e.g. the color dye, matte surface, cloth)
  - perceived color depends on reflected freq (=emitted freq-absorbed freq.)
  - follows subtractive rule
    - R+G+B=Black

Light and Spectra

- Spectral Power Distribution (SPD) or a spectrum showing the relative power for typical outdoor light on a sunny day.

$$E(x, y, z, \lambda)$$

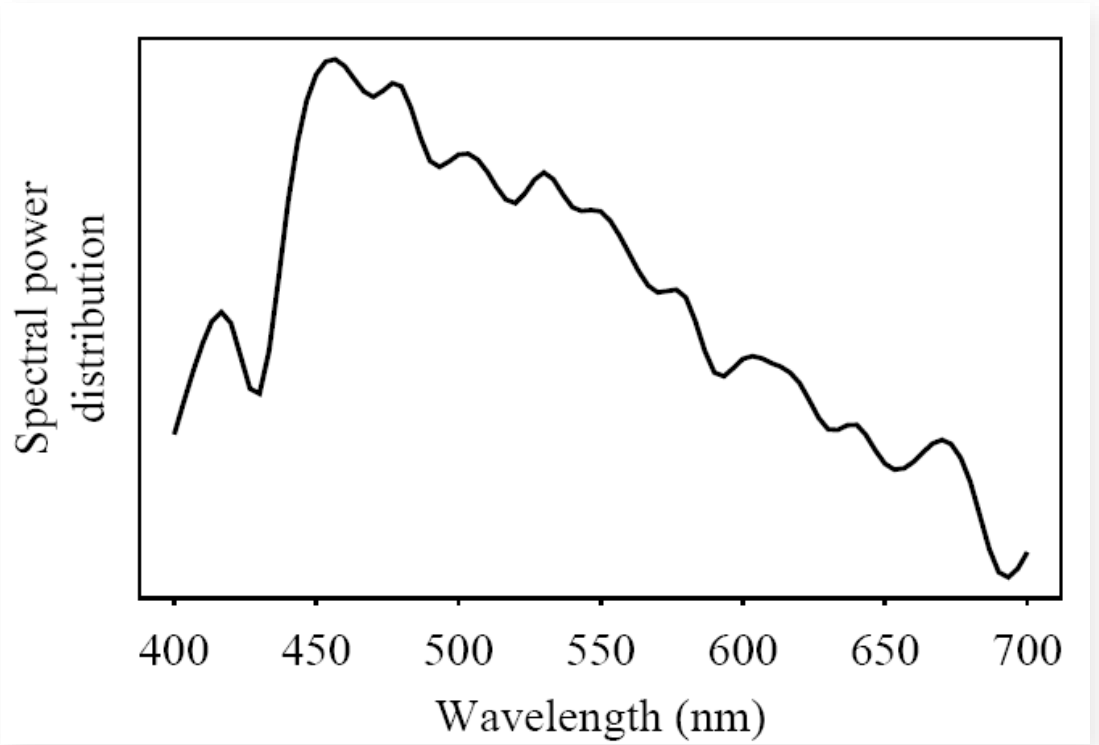
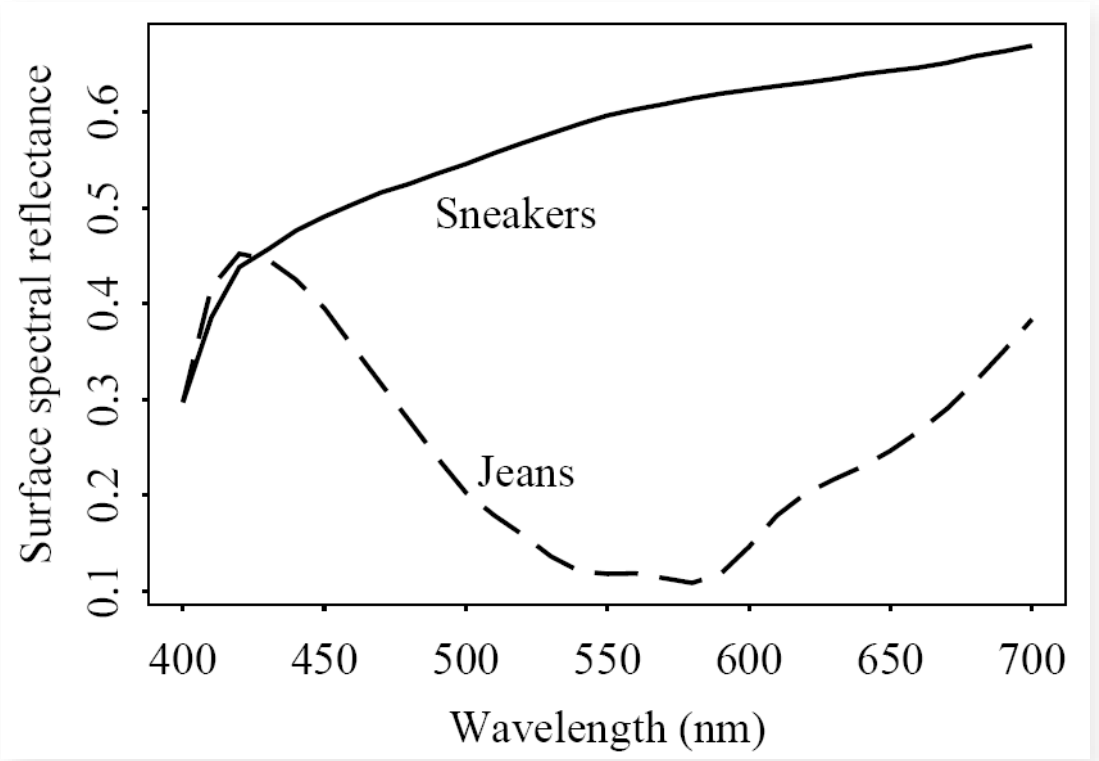


Image Formation

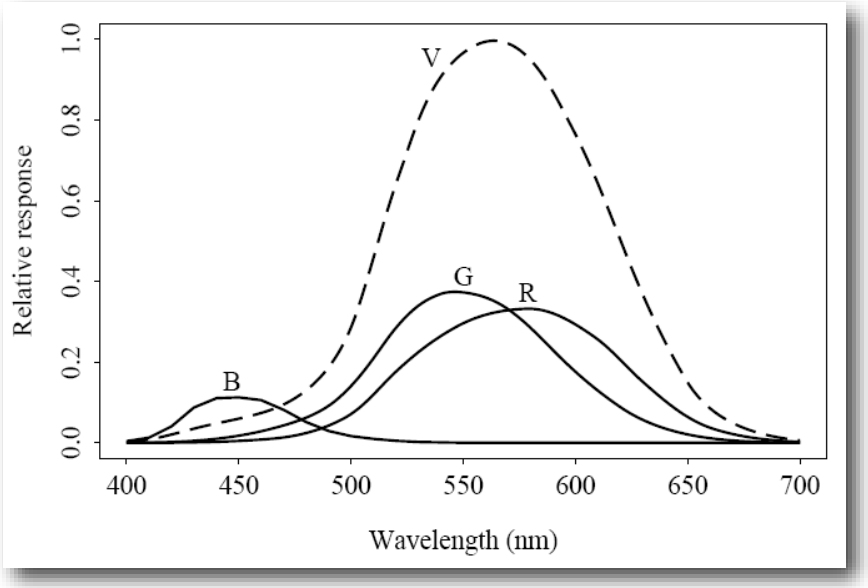
- Surfaces reflect different amounts of light at different λ
- Dark surfaces reflect less light energy than light surfaces
- The amount of reflected light for a specific surface is described by the reflectance function

$$R(x, y, z, \lambda)$$



Spectral Sensitivity of the Eye

- The eye is most sensitive to light in the middle of the visible spectrum.
- The sensitivity of our receptors is also a function of wavelength.
- The Blue receptor sensitivity is not shown to scale because it is much smaller than the curves for Red or Green.



- V(λ) is the overall luminous-efficiency sensitivity function

Image capture summary

- For natural images we need a light source

$$E(x, y, z, \lambda)$$

- Each point in the scene has a reflectivity function

$$R(x, y, z, \lambda)$$

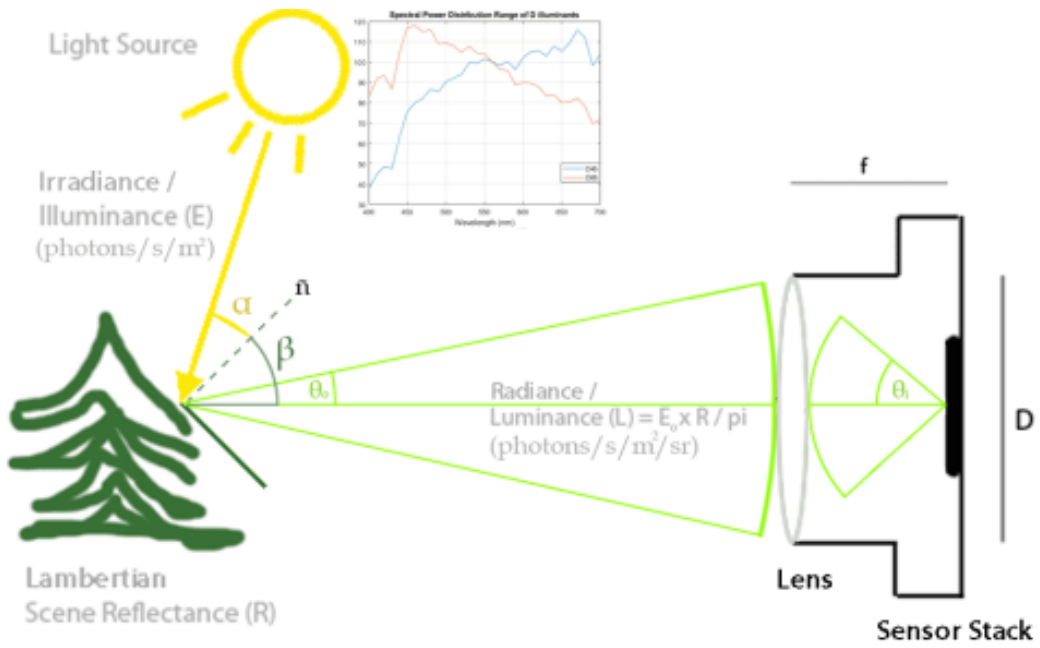
- Light reflects from a point and the reflected light is captured by an imaging device

$$c(\boldsymbol{X}, \lambda) = E(\boldsymbol{X}, \lambda) \cdot R(\boldsymbol{X}, \lambda)$$

- Perspective projection from space to image

$$\mathbf{X} \underset{P}{\rightarrow} \mathbf{x}$$

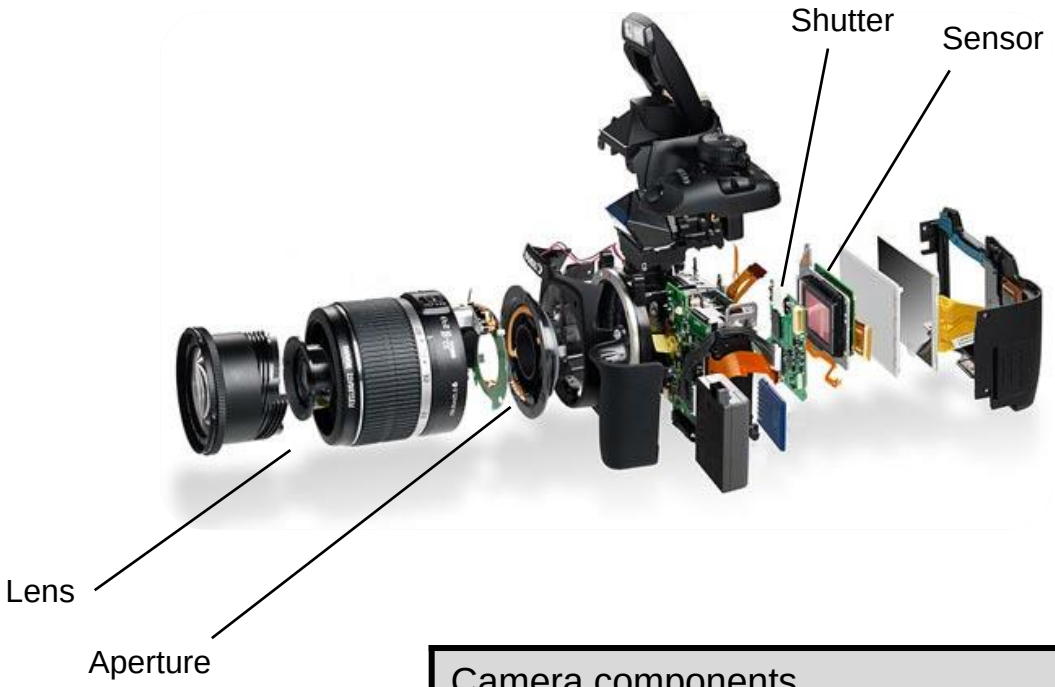
$$\psi(P(\mathbf{X})) = \bar{\psi}(\mathbf{X}) \quad \text{or} \quad \psi(\mathbf{x}) = \bar{\psi}(P^{-1}(\mathbf{x}))$$

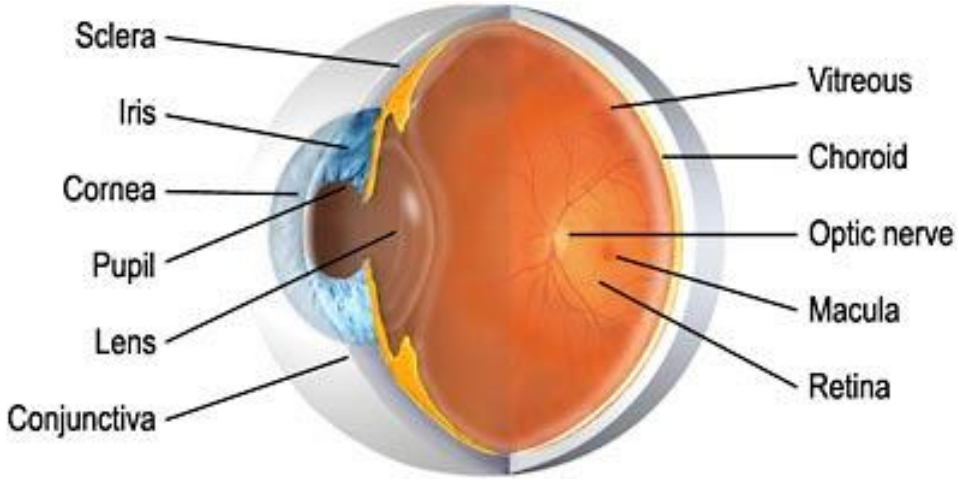


- The camera sensor has an absorbtion function

$$\bar{\psi}(\mathbf{X}) = \int C(\mathbf{X}, \lambda)\, a_c(\lambda) d\lambda$$

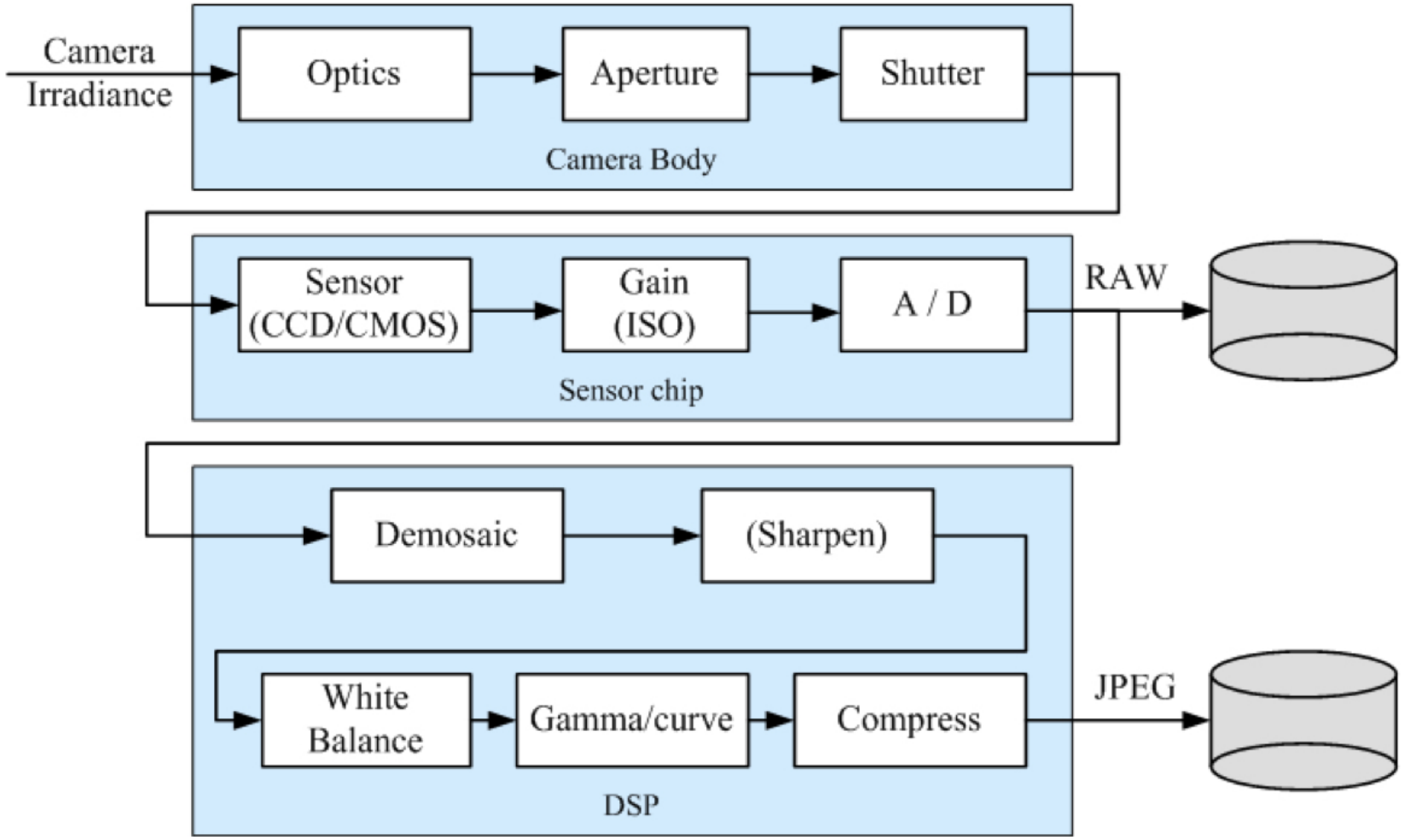
Eye vs. Camera





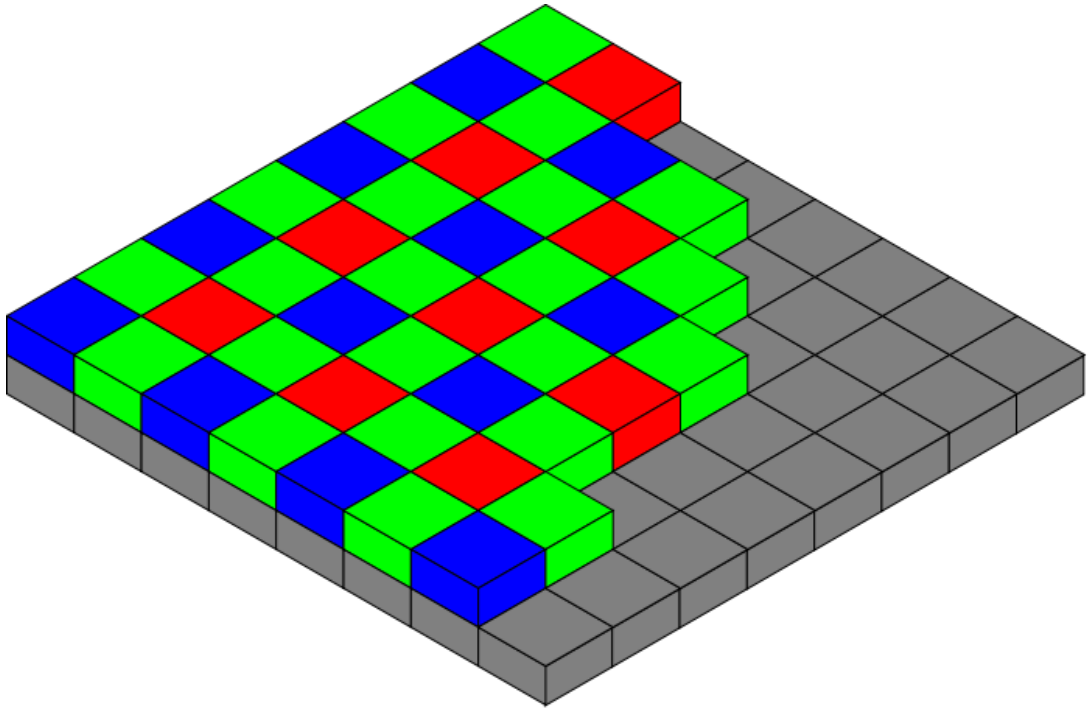
Camera components	Eye components
Lens	Lens, cornea
Shutter, aperture	Iris, pupil
Sensor or film	Retina
Cable to transfer images	Optic nerve to send visual stimuli to the brain

Digital camera



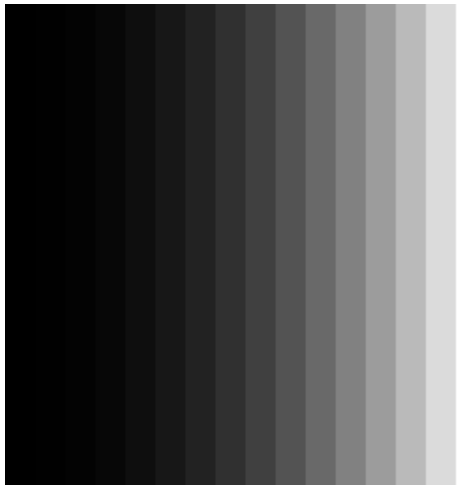
Camera Systems

- The analog signal from the image sensor is converted to digital form
- A color filter converts a pixel to either R, G, or B.
- Interpolation is used to give each pixel all three color components

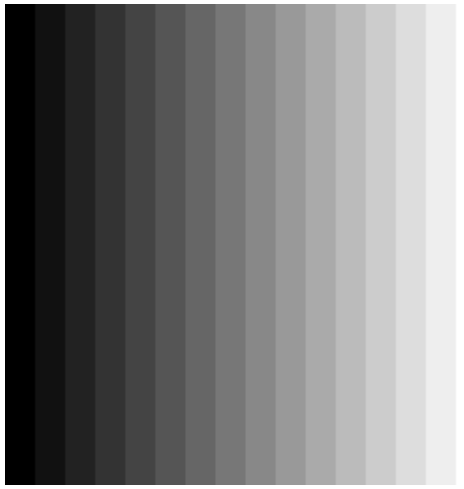


Gamma Correction

- When viewing a captured image the R, G, and B color values are used to illuminate a pixel on the display
- The display may not have a linear relationship between color value and produced light



No Gamma correction

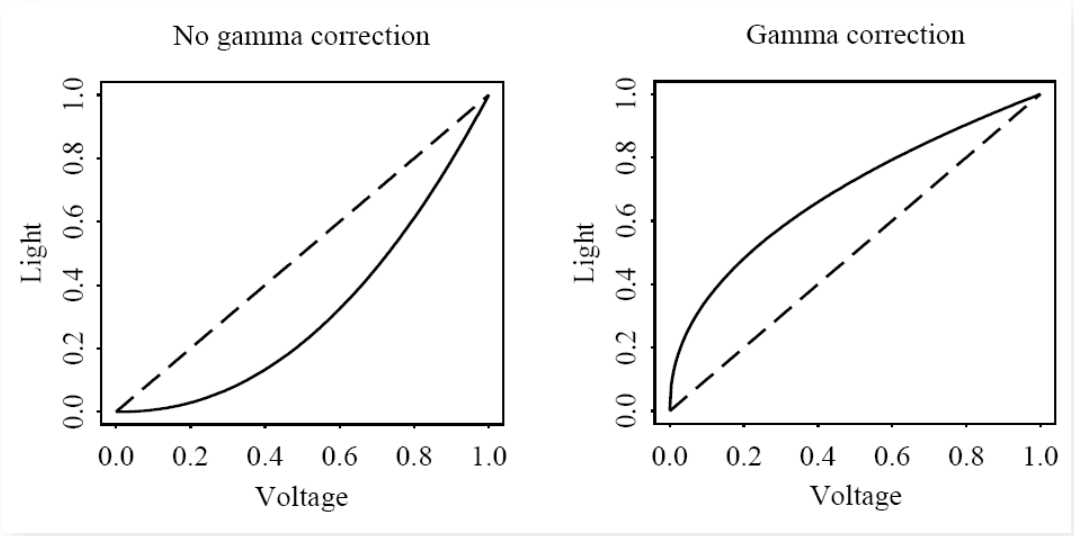


With Gamma correction

Gamma Correction

Historically cameras have pre-corrected for this nonlinearity by applying the inverse to the pixel values: gamma correction

$$C \rightarrow C' = C^{1/\gamma} \Rightarrow (C')^{\gamma} \rightarrow C$$



CIE Chromaticity Diagram

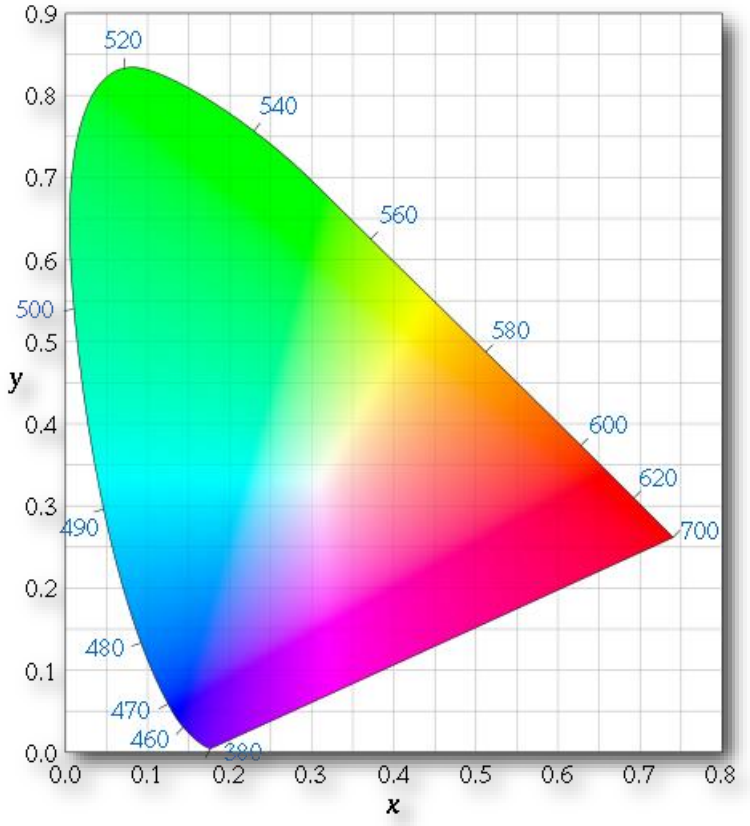
$$x = X/(X + Y + Z)$$

$$y = Y/(X + Y + Z)$$

$$z = Z/(X + Y + Z)$$

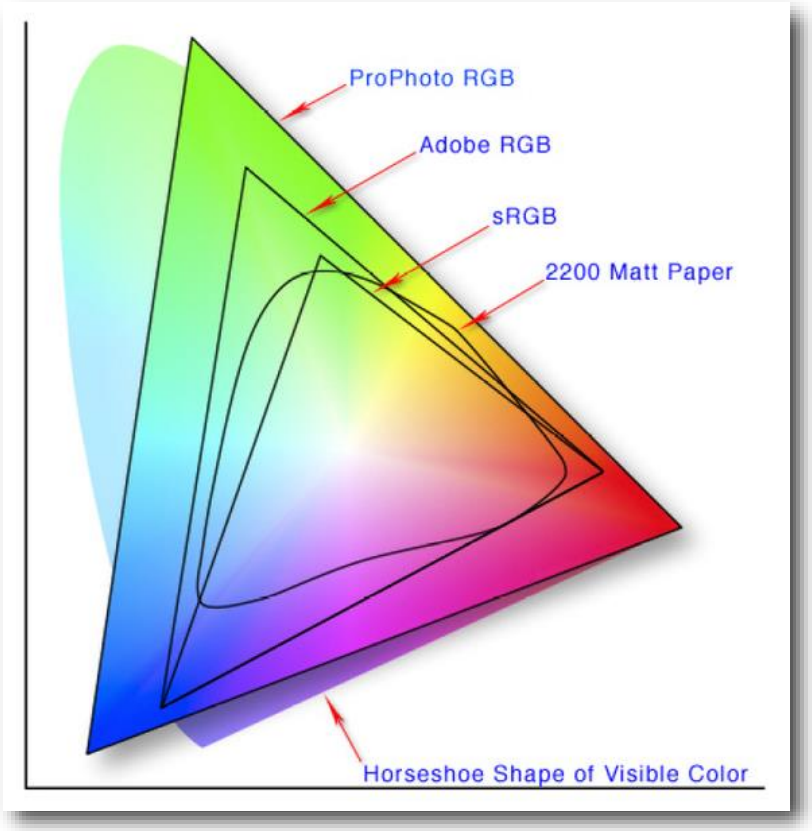
$$x + y + z = \frac{X + Y + Z}{X + Y + Z} \equiv 1$$

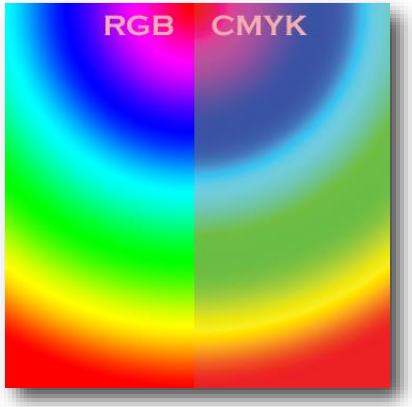
Each 3D vector $[X,Y,Z]$ is effectively projected down to the 2D plane, which connects points (1, 0, 0), (0, 1, 0), and (0, 0, 1).



Color space

- Each medium spans its own colorspace within which it can represent colors
- The color gamut (range) might differ significantly

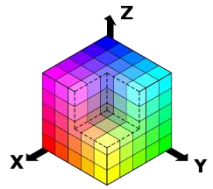




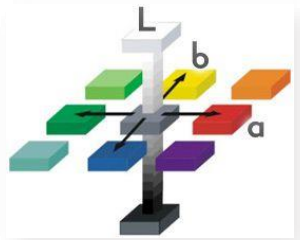
Display Paper

Color Models

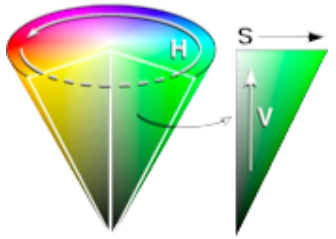
- RGB



- CMYK (Cyan, Magenta, Yellow, Key = Black)
- LAB

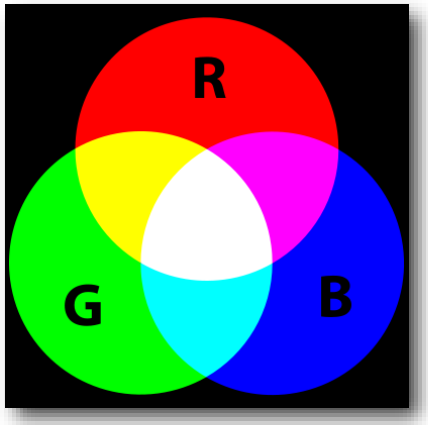


- YUV, YCbCr (Y - Luminance, the other chromacity)
- HSV (Hue, Saturation, Value)

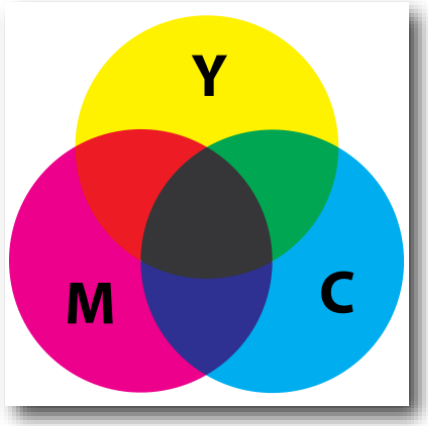


Color Models

- Additive
  Light combine into a color

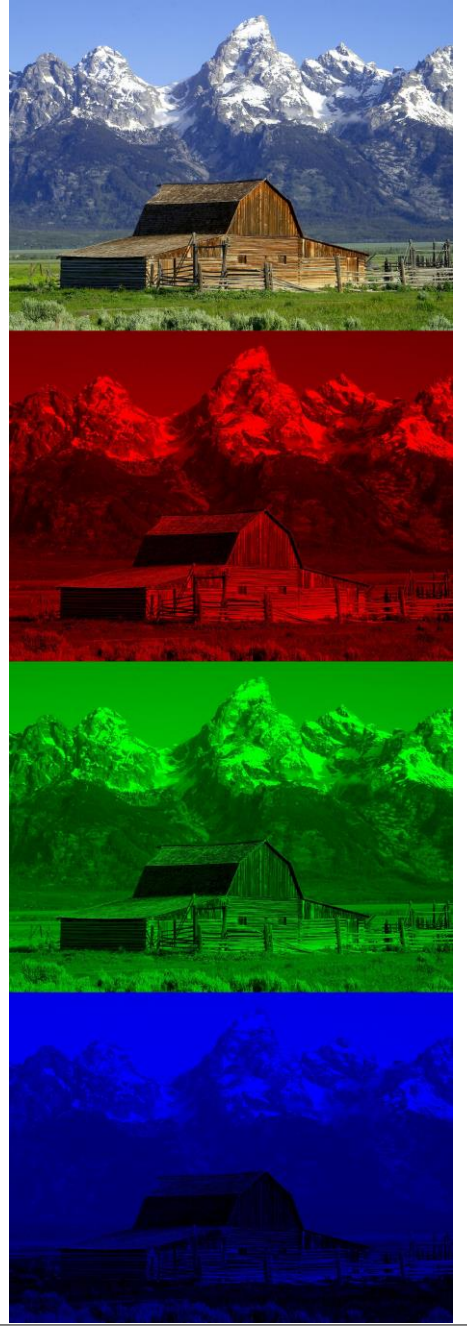


- Subtractive
  Ink filters incident light into a resulting color



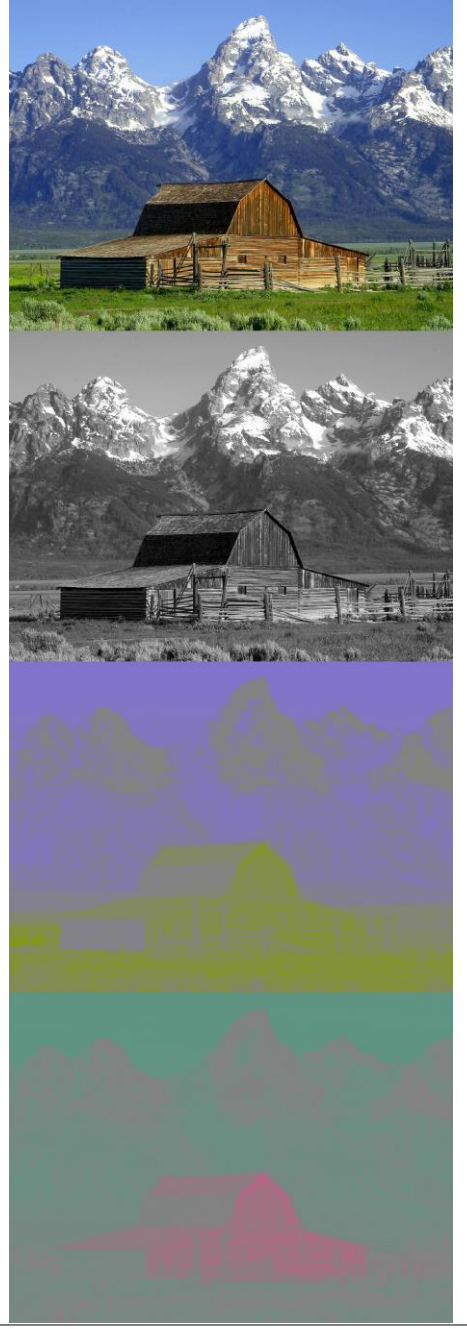
RGB

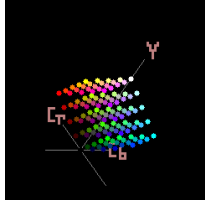
- Additive Color Model
  - Colors are created by adding light in the Red, Green, and Blue channels
  - Basis of digital imaging and display technology (monitors, projectors, etc.)
- Channel Separation
  - Each pixel is defined by a triplet: (R, G, B), and each channel controls the intensity of light at that color
- Human Vision Simulation
  - Mimics the way cones in the human eye detect light.
- Limitations
  - Not perceptually uniform - equal changes in RGB values do not correspond to equal perceived changes in color.
  - Limited color gamut - doesn't cover all visible colors, leading to the development of other color models.



YCbCr

- Luminance and Chrominance Model
  - Separates an image into luminance (Y) and chrominance (Cb, Cr) components.
  - Y represents brightness or grayscale information, while Cb and Cr represent color differences (blue-yellow and red-green).
- Human Vision Alignment
  - The Y channel aligns with the human eye's higher sensitivity to brightness than color differences.
  - Often utilizes 4:2:0 chroma subsampling to compress data by reducing chrominance resolution without significant perceptual loss.
- Applications
  - Widely used in video encoding, broadcasting, and image processing for compression and efficient transmission.





Color Model transformation

From RGB to Y'CbCr

$$\begin{bmatrix} Y' \\ C_b \\ C_r \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.168736 & -0.331264 & 0.5 \\ 0.5 & -0.418688 & -0.081312 \end{bmatrix} \begin{bmatrix} R' \\ G' \\ B' \end{bmatrix} + \begin{bmatrix} 0 \\ 0.5 \\ 0.5 \end{bmatrix}$$

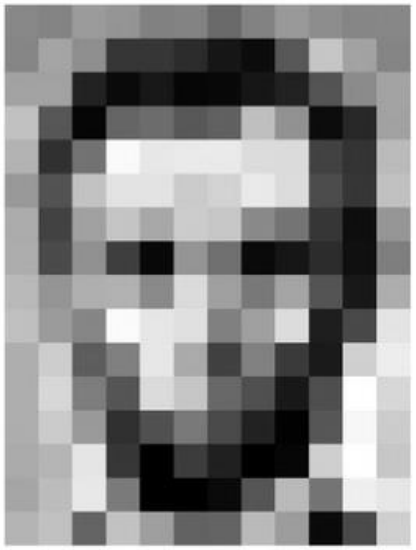
8-bit coding of the color components gives

$$\begin{bmatrix} Y' \\ C_b \\ C_r \end{bmatrix} = \begin{bmatrix} 65.481 & 128.553 & 24.966 \\ -37.797 & -74.203 & 112 \\ 112 & -93.786 & -18.214 \end{bmatrix} \begin{bmatrix} R' \\ G' \\ B' \end{bmatrix} + \begin{bmatrix} 16 \\ 128 \\ 128 \end{bmatrix}$$

Y'CbCr transform is used in JPEG image and MPEG video compression.

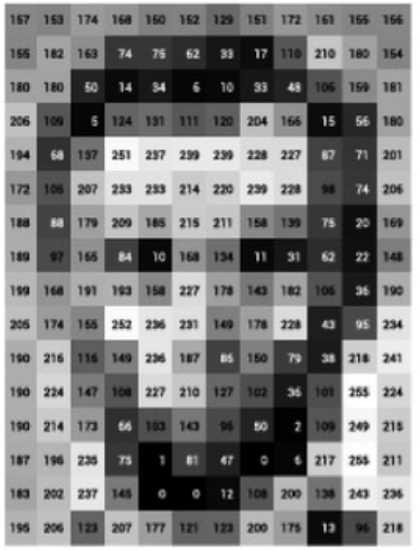
Image as a function, matrix, pixel map, …

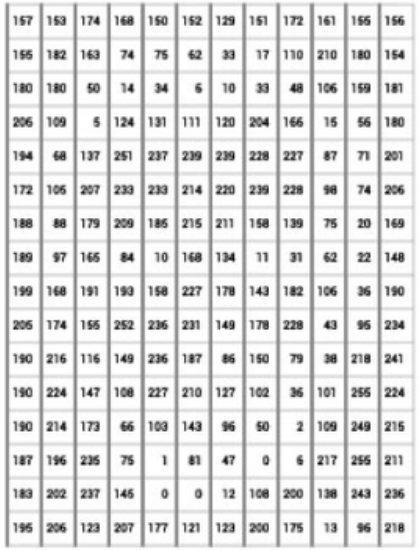
$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(M-1,N-1) \\ \vdots & \vdots & \ddots & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$



where

$$0 \leq f(x,y) \leq G - 1$$





- Usually G are expressed as integer powers of 2 ($G = 2^n$)
- For simplicity often $N = M$

Image as a function, matrix, pixel map, …

$\boldsymbol{I} = f(\boldsymbol{x}), where\ \boldsymbol{x} \in \mathbb{R}^2$

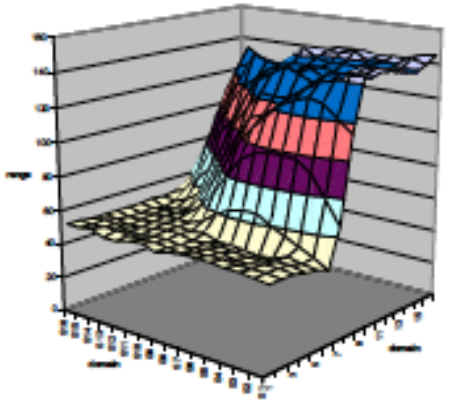
$\boldsymbol{I} \in \mathbb{R}$ - *grayscale*

$\boldsymbol{I} \in \mathbb{R}^3$ - *color*

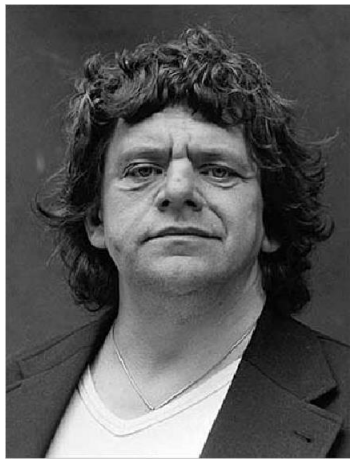
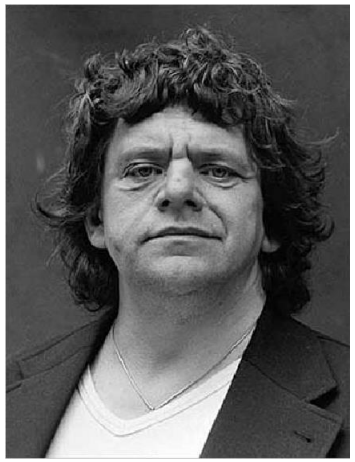
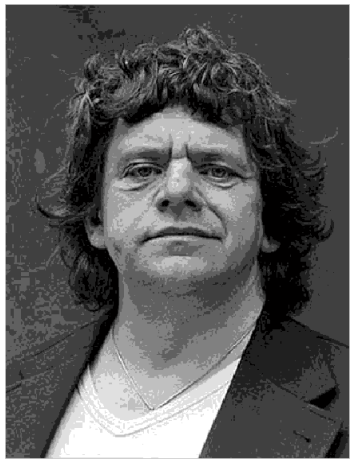
Conventionally images are definied over a discreete domain with a range of finite precision

$\boldsymbol{I} = f(\boldsymbol{x}), where\ \boldsymbol{x} \in \mathbb{N}^2$
$\boldsymbol{I} \in \mathbb{N}\ or\ \boldsymbol{I} \in \mathbb{N}^3$

45	60	98	127	132	133	137	133
46	65	98	123	126	128	131	133
47	65	96	115	119	123	135	137
47	63	91	107	113	122	138	134
50	59	80	97	110	123	133	134
49	53	68	83	97	113	128	133
50	50	58	70	84	102	116	126
50	50	52	58	69	86	101	120



Discrete images - sampling and quantization

$f_s \cong 2f_{max}$	$\frac{f_s}{2}$	$\frac{f_s}{4}$	$\frac{f_s}{10}$
			
			
$8\ bpp$ $256\ gray\ levels$	$4\ bpp$ $16\ gray\ levels$	$2\ bpp$ $4\ gray\ levels$	$1\ bpp$ $2\ gray\ levels$

How much space does an image take?

- Grayscale image

$$b = M \cdot N \cdot m \; [bits]$$

$$r = \frac{b}{M \cdot N} = m \; [bits/pixel] = [bpp]$$





- Color image (Red, Green, Blue)

$$b = M \cdot N \cdot 3m \; [bits]$$

$$r = \frac{b}{M \cdot N} = 3m \; [bpp]$$



Image representation scheme

- Different representation formats serve different purposes

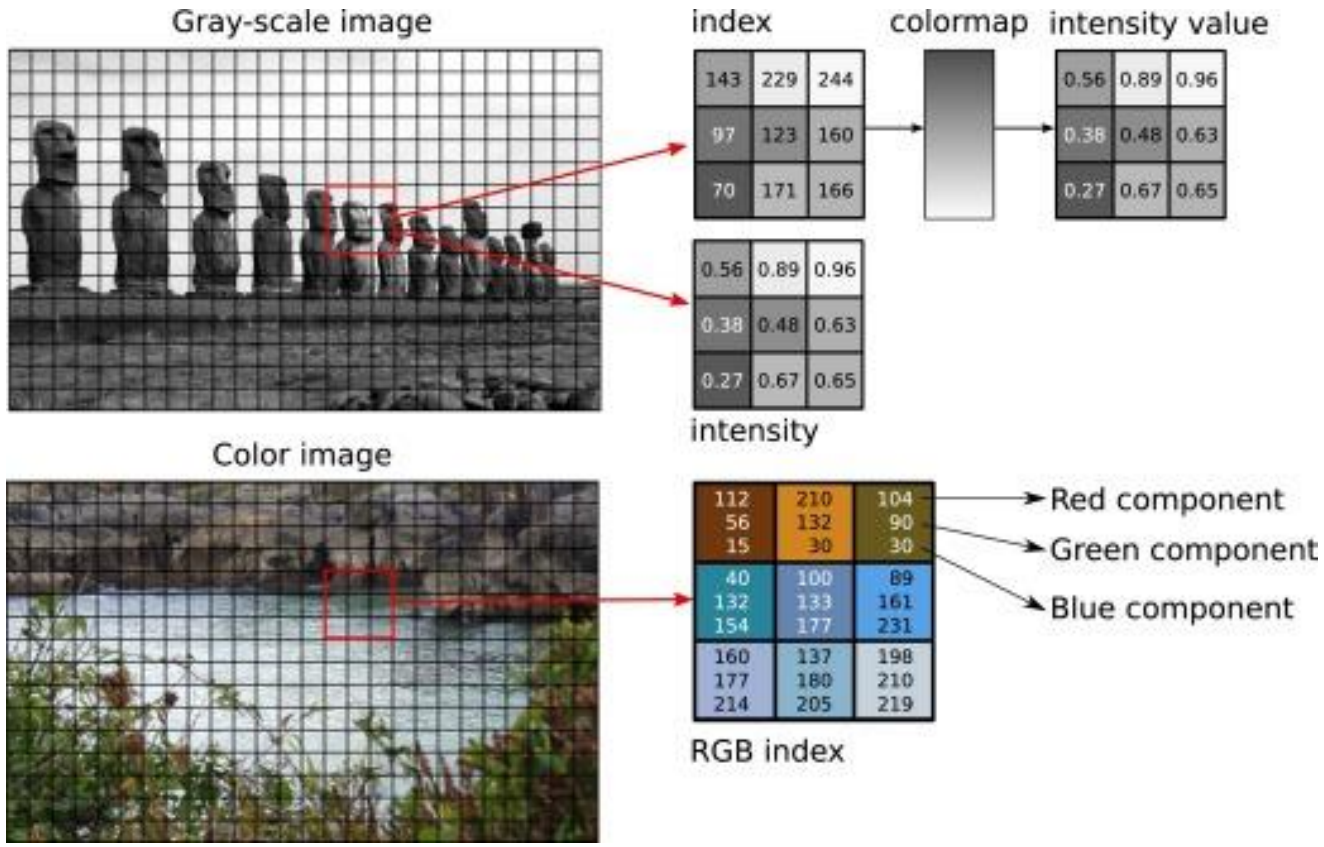
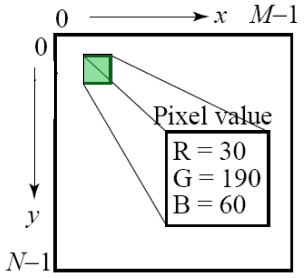


Image examples

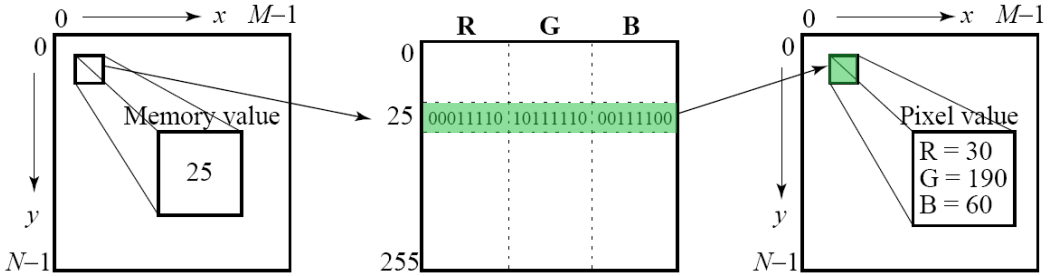
Bitplane based images

The value at each pixel either:

- holds a representation of a specific color



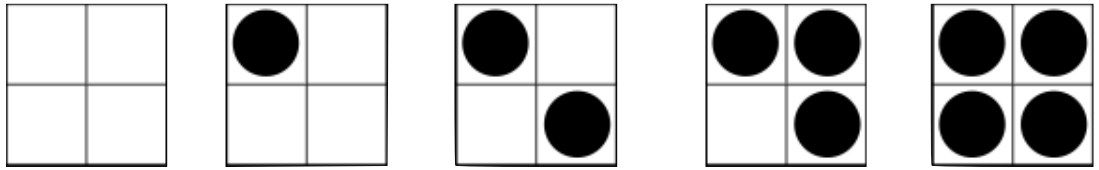
- hold an index to a palette, which defines a set of colors



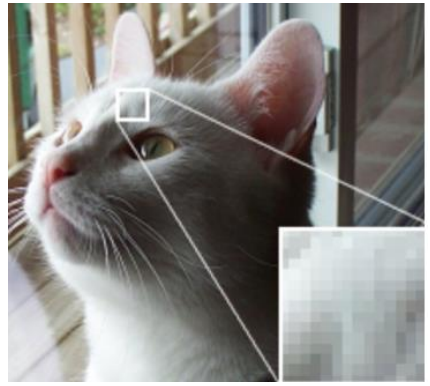
If the number of colors used are $\ll M \cdot N$ this results in significant space reduction

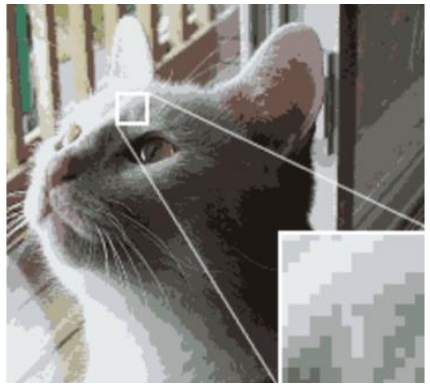
Dithering

- Dithering can make the number of colors *seem* greater.
- Spatial resolution is traded for increased intensity resolution.
- A pixel value is replaced by a specific pattern, with an average color corresponding to the original pixel.

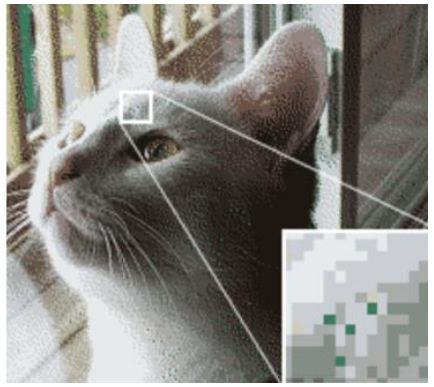


A pattern with size 2x2 allows for five gray levels





$G = 16$



$G = 16$
with dither